

Screenshots

1 Data Loading Output

refGene	SGIP1	NECAP2	AZIN2	AGBL4	CLIC4	SLC45A1	TGFBR3	DBT	PRUNE1	C1orf21	...	MGC70870	KIR2DL5A	KIR2DS2	KIR2DL2	KIR3DS1	KIR2DS3	KIR2
20-1T-AD	1405	295	356	191	876	291	639	623	297	2456	...	26	0	0	0	0	0	
13-11T-Old	1405	460	306	200	1443	329	650	726	237	2006	...	57	0	0	0	0	0	
15-13T-Old	1169	334	385	173	639	298	506	513	200	2092	...	25	0	0	0	0	0	
16-14T-Old	2408	347	507	323	792	636	370	633	390	3146	...	36	0	0	0	0	0	
3-17T-Young	859	617	787	36	4806	139	425	751	231	848	...	109	0	0	0	0	0	

5 rows x 27136 columns

The dataset was loaded and transposed successfully. It contains **30 rows (samples)** and **27,136+ columns (genes)** plus the Target column. Each row represents one biological sample (AD or Control), and each column represents the raw read count of a gene. The head output confirms the correct structure after transposition.

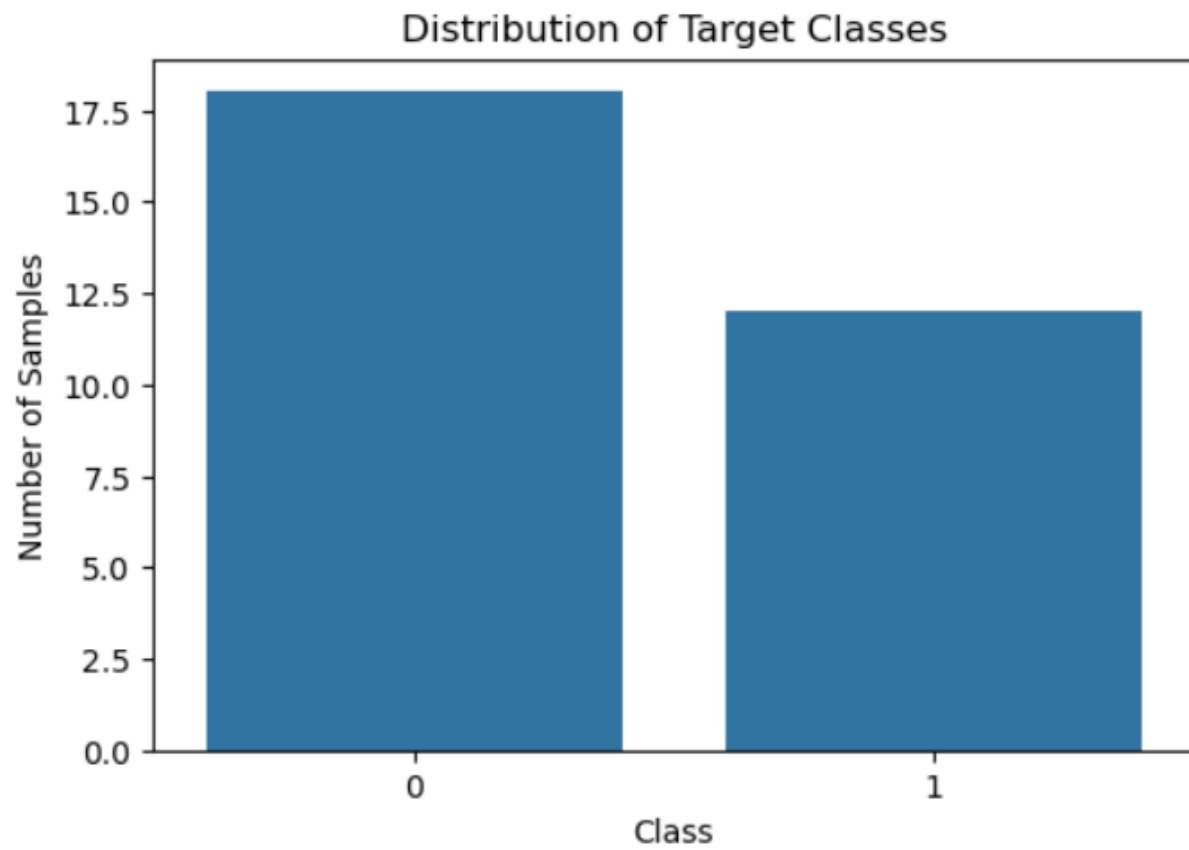
2 Summary Statistics

refGene	SGIP1	NECAP2	AZIN2	AGBL4	CLIC4	SLC45A1	TGFBR3	DBT	PRUNE1	C1orf21	...	MGC70870	KIR2
count	30.000000	30.000000	30.000000	30.000000	30.000000	30.000000	30.000000	30.000000	30.000000	30.000000	...	30.000000	
mean	1560.400000	382.633333	396.633333	188.966667	1893.233333	284.333333	485.833333	655.266667	277.333333	2142.500000	...	50.800000	
std	665.418437	98.389964	152.339567	99.354289	1891.945554	129.955253	144.322456	178.788631	58.104741	720.807677	...	41.801996	
min	795.000000	196.000000	179.000000	36.000000	639.000000	138.000000	172.000000	397.000000	188.000000	525.000000	...	5.000000	
25%	1097.250000	339.250000	297.250000	123.250000	828.250000	193.500000	393.500000	513.250000	232.000000	1833.250000	...	24.250000	
50%	1363.500000	355.500000	372.500000	187.500000	1251.000000	252.000000	481.000000	644.000000	262.000000	2090.000000	...	37.500000	
75%	1714.500000	425.500000	436.750000	243.000000	1505.250000	344.000000	574.750000	754.750000	305.750000	2451.750000	...	62.250000	
max	3115.000000	617.000000	787.000000	431.000000	8812.000000	636.000000	792.000000	1050.000000	411.000000	3800.000000	...	182.000000	

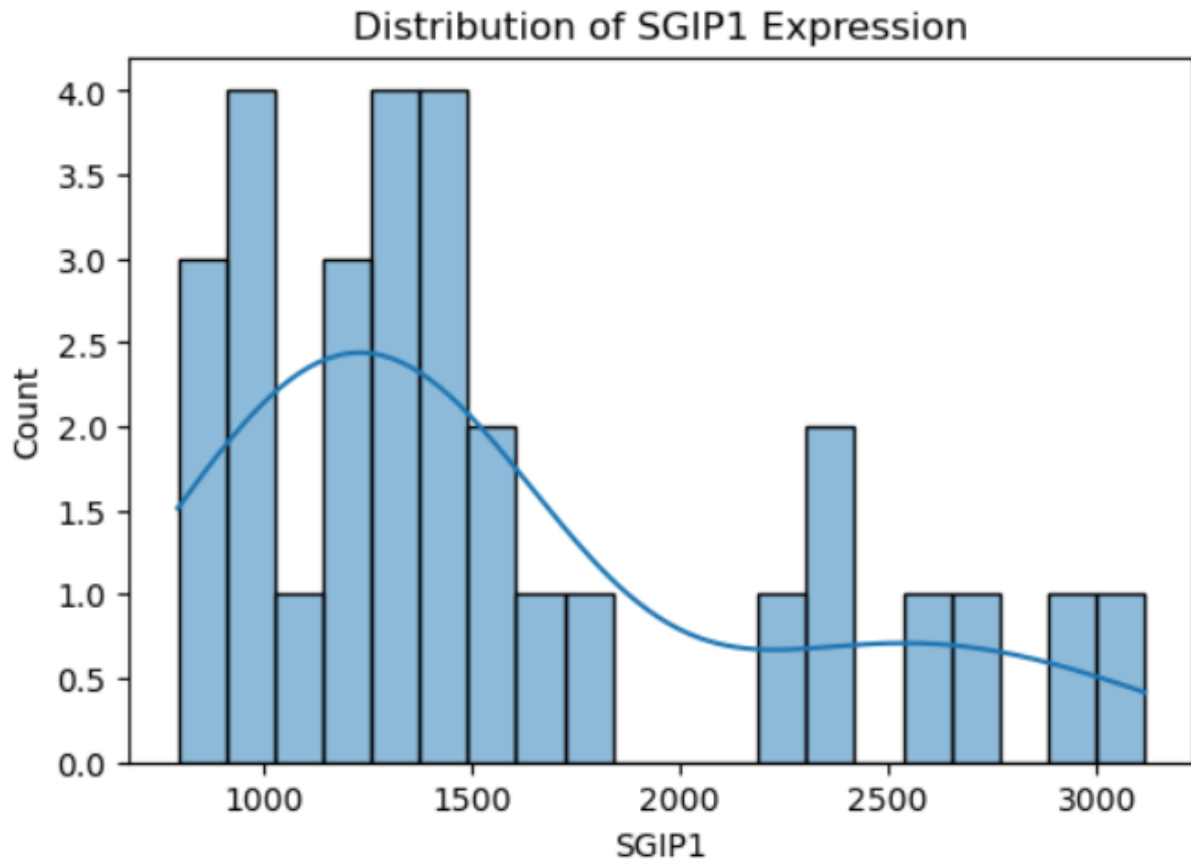
1 rows x 27136 columns

The summary statistics reveal that gene expression values vary widely across features. Many genes have a mean close to zero with very high standard deviation, indicating that most genes are lowly expressed in most samples while a few genes have very high counts. This right-skewed distribution is typical of raw RNA-seq count data and supports the need for log transformation or feature selection before modeling

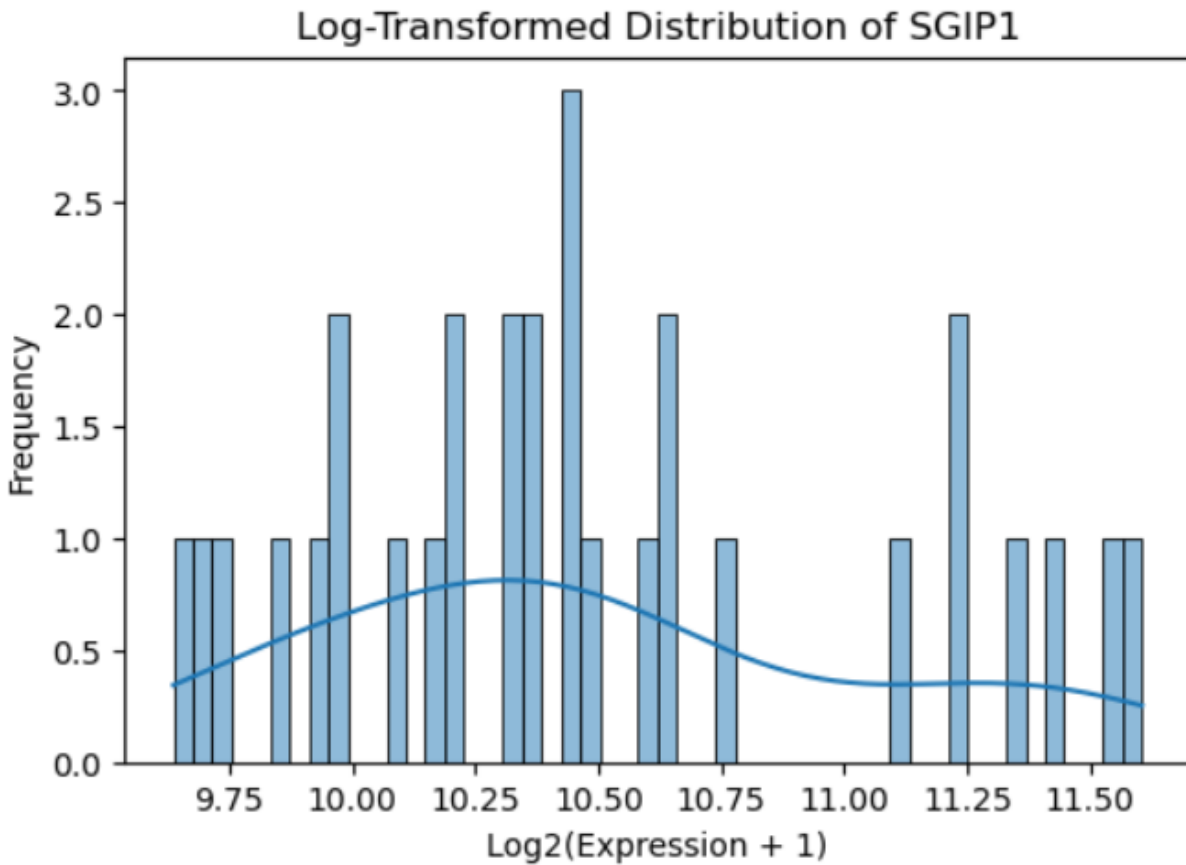
3 countplot — Target Variable Distribution



4 histplot — SGIP1 Raw Distribution



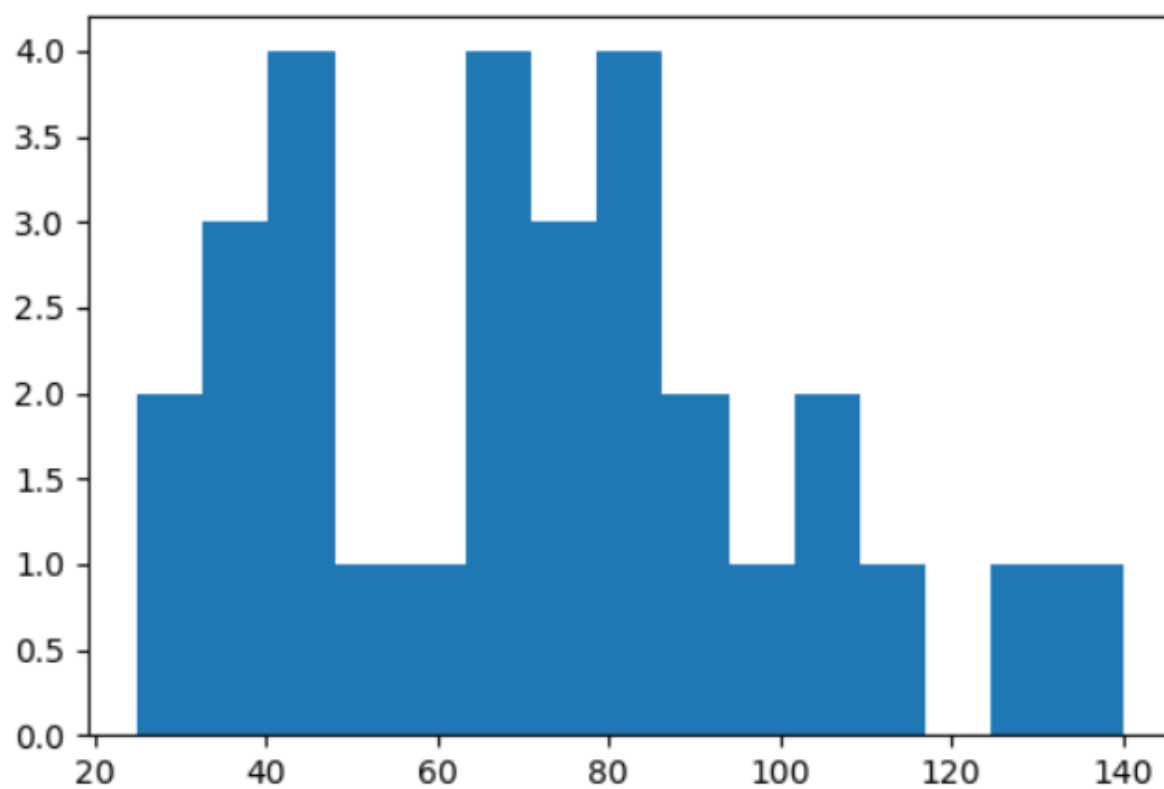
The raw expression distribution of SGIP1 is strongly right-skewed. The majority of samples cluster near zero, while a small number of samples show extremely high expression values. This pattern is characteristic of RNA-seq count data where most genes are not highly expressed across all samples.

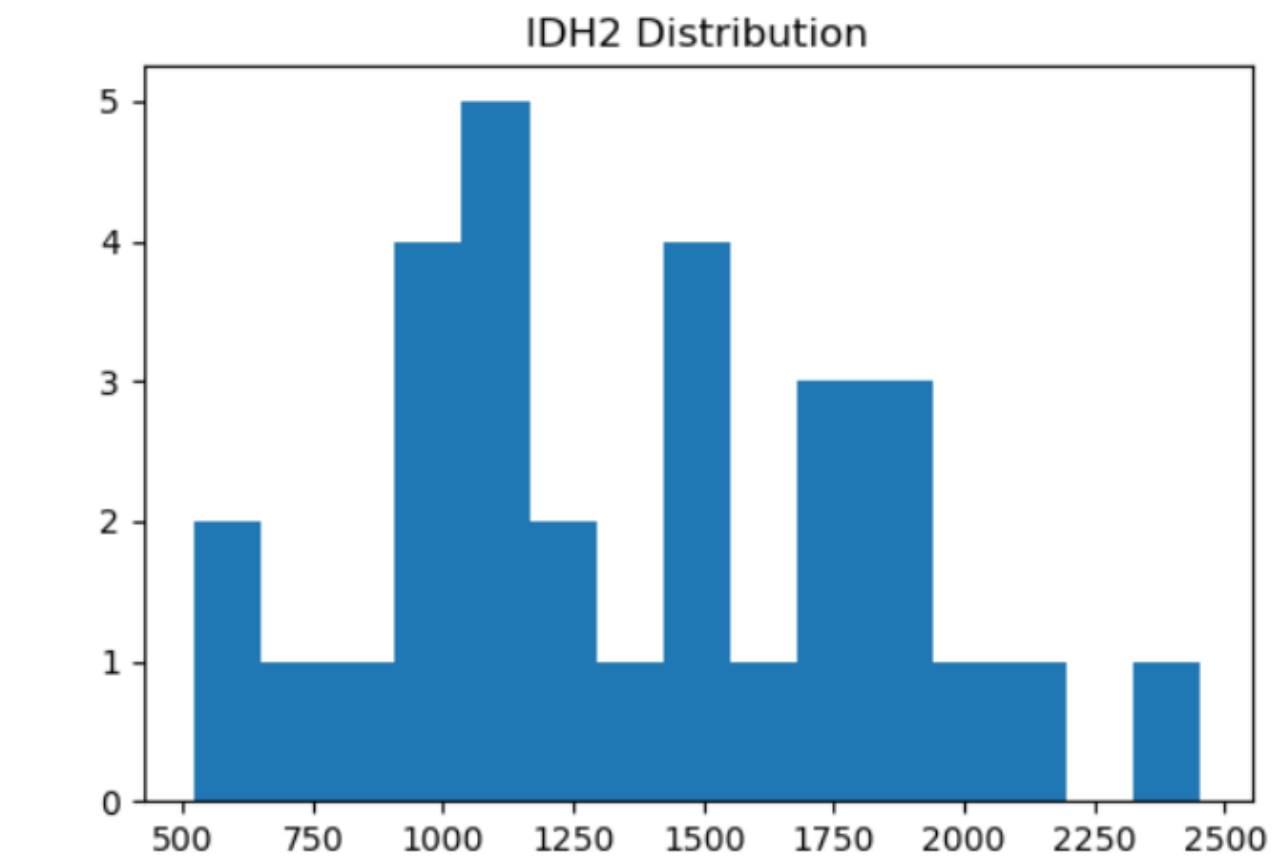


After applying $\log_2(x+1)$ transformation, the distribution of SGIP1 becomes more symmetric and approximately normal. This confirms that the raw count distribution was highly skewed and that log transformation is an appropriate preprocessing step for visualizing and interpreting RNA-seq data.

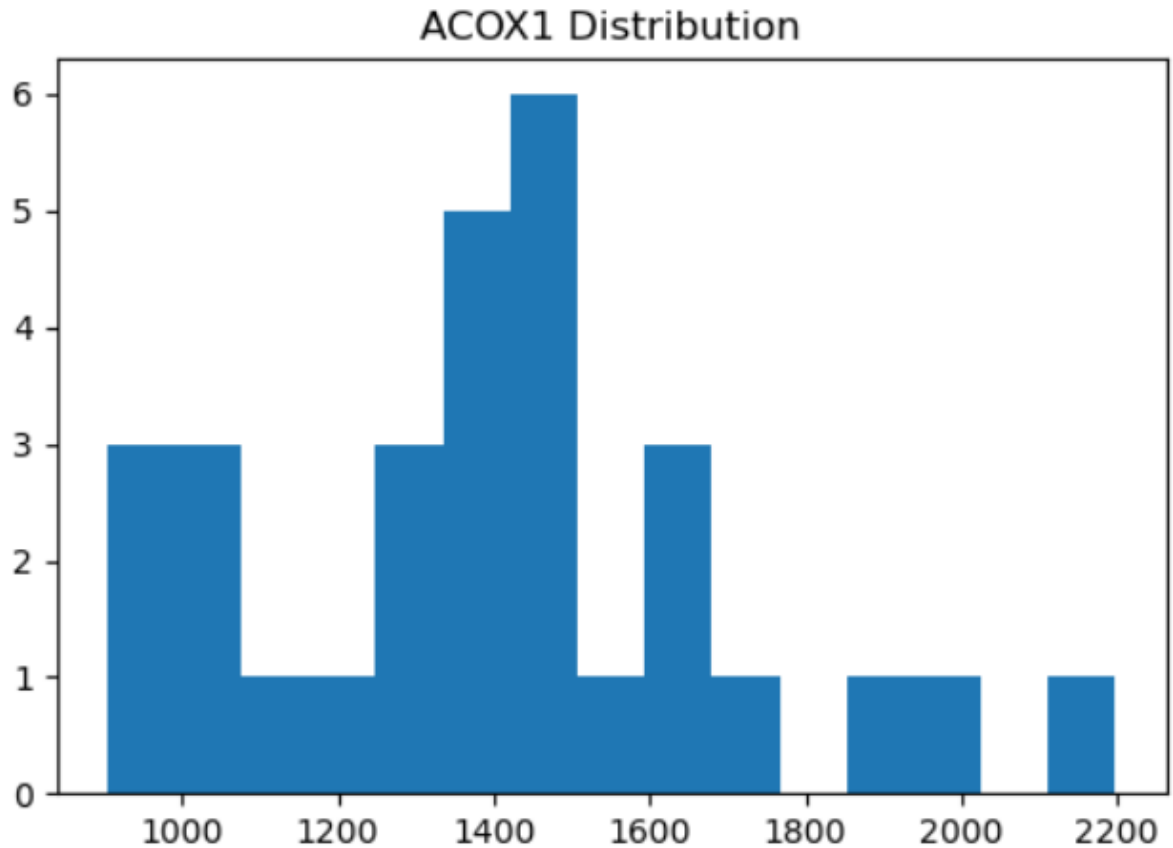
`plt.hist()` Loop — ERI3-IT1, IDH2, ACOX1 Distributions

ERI3-IT1 Distribution





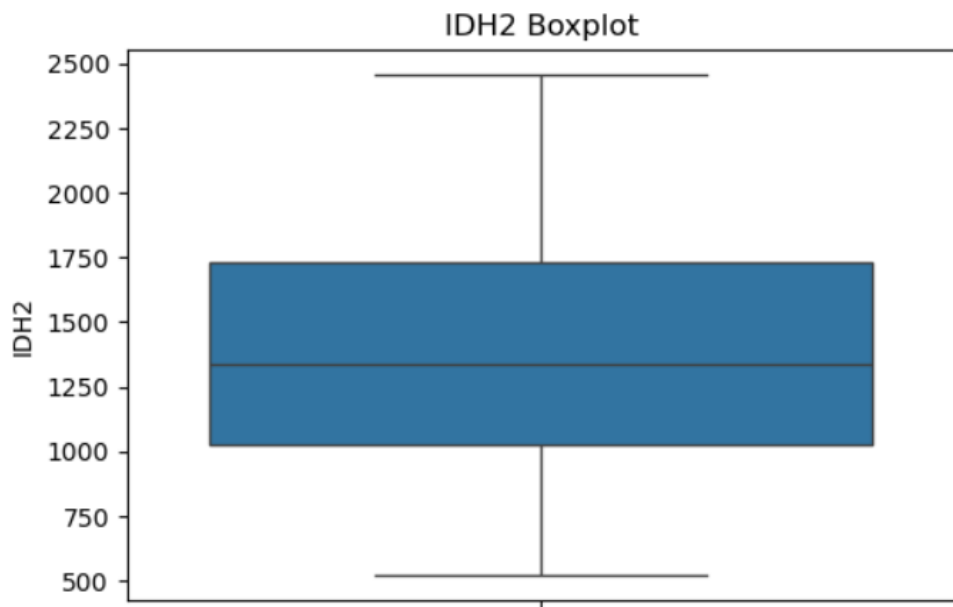
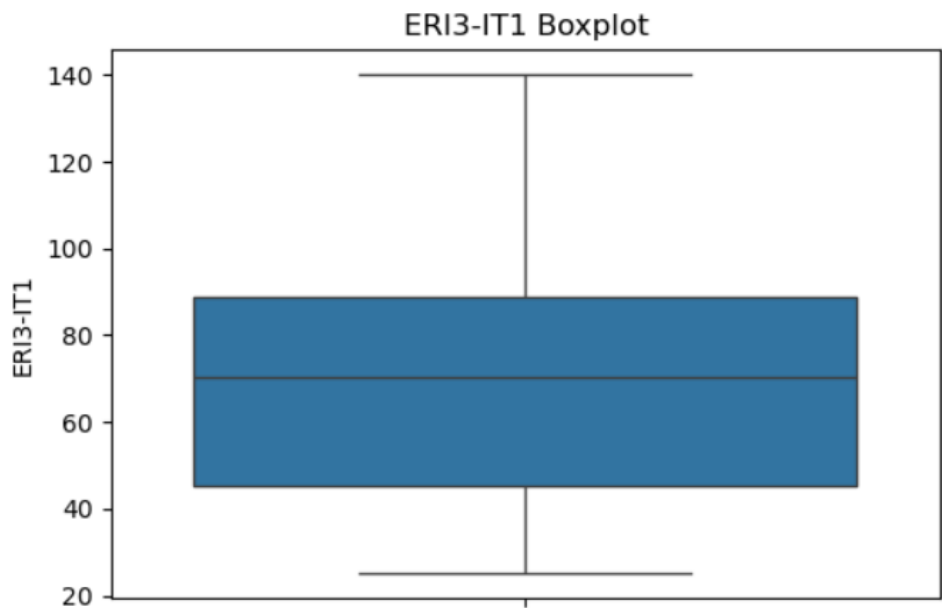
.

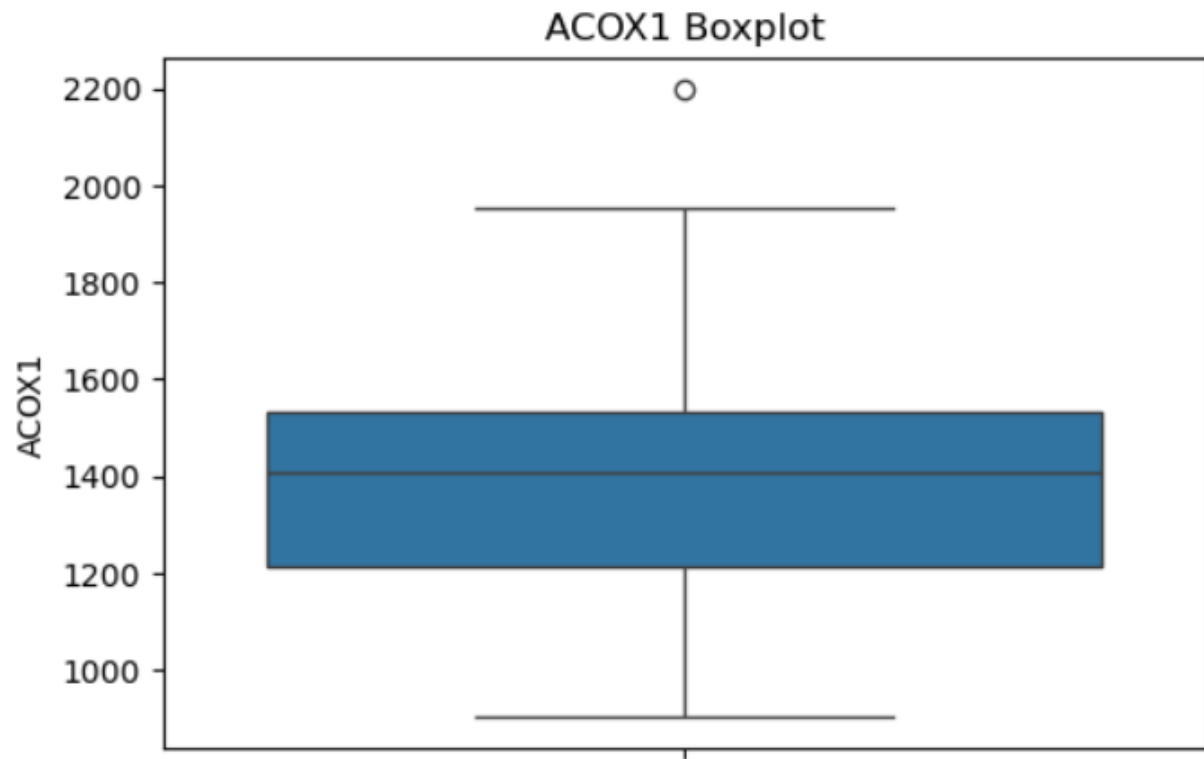


ERI3-IT1: Shows a highly skewed distribution, with most samples having near-zero expression. A few samples show elevated counts, suggesting this gene may be differentially expressed in a subset of AD samples.

- **IDH2:** The distribution is more spread, with moderate expression across most samples. The variation across samples suggests IDH2 could carry discriminative signal.
- **ACOX1:** Similar right-skewed pattern. The spread of values between samples makes it a candidate for distinguishing disease from control groups.

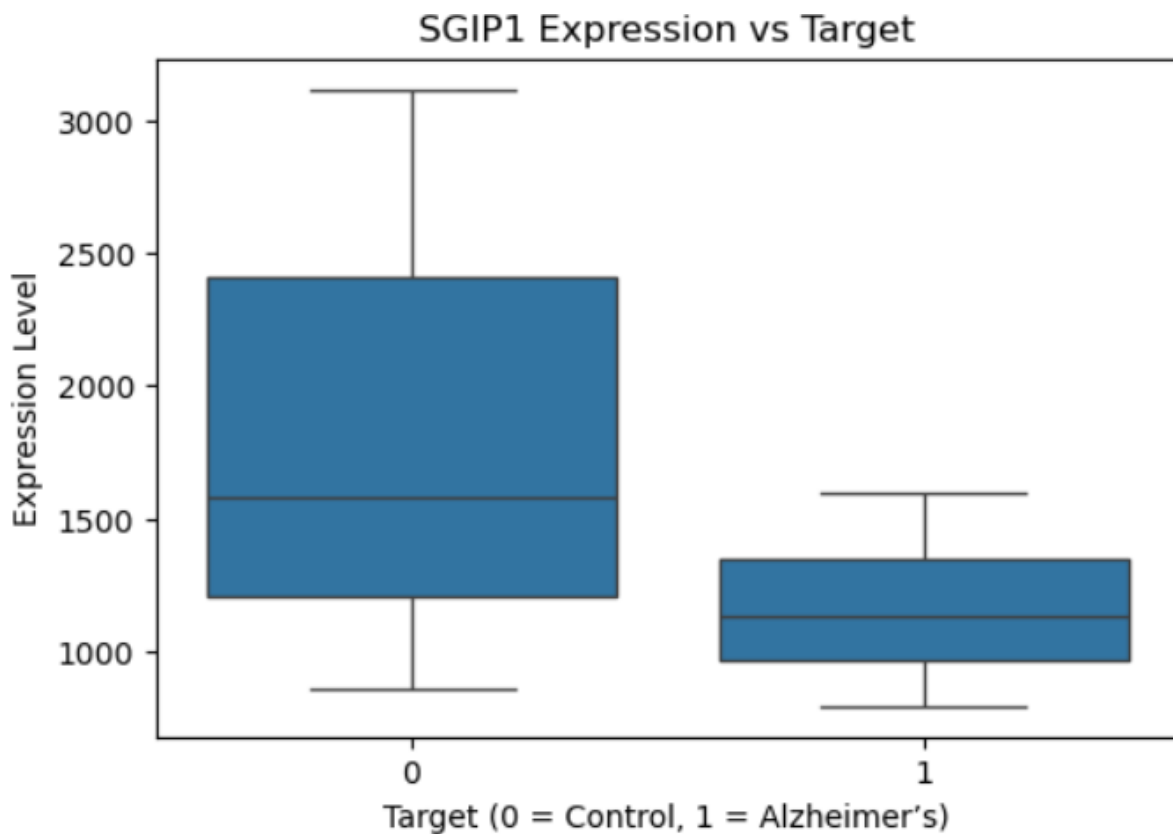
boxplot Loop — ERI3-IT1, IDH2, ACOX1





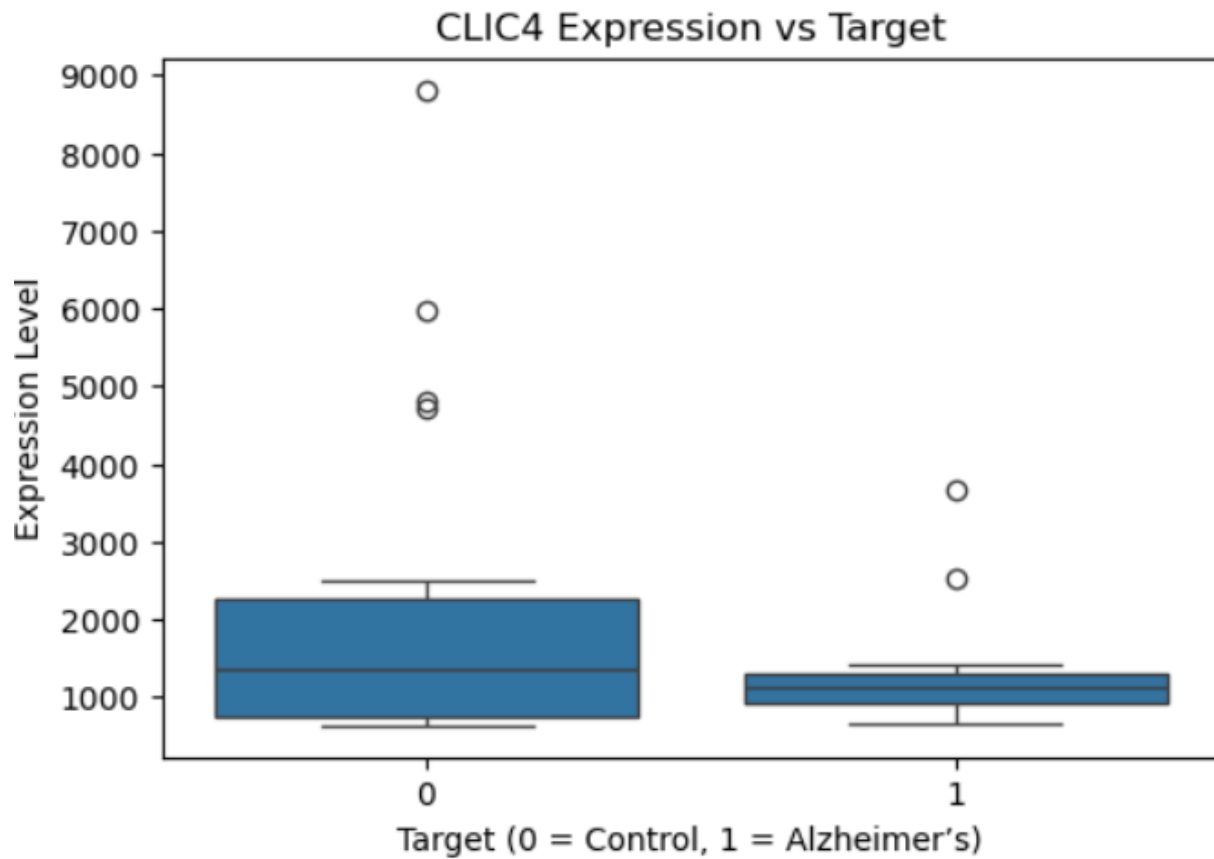
These boxplots show the spread and outliers for each gene across all samples. Genes with wide interquartile ranges (IQR) and many outlier points contain high variability — which is informative for classification. Genes with a tight IQR and few outliers are more uniformly expressed and may be less useful as features.

boxplot — SGIP1 vs Target



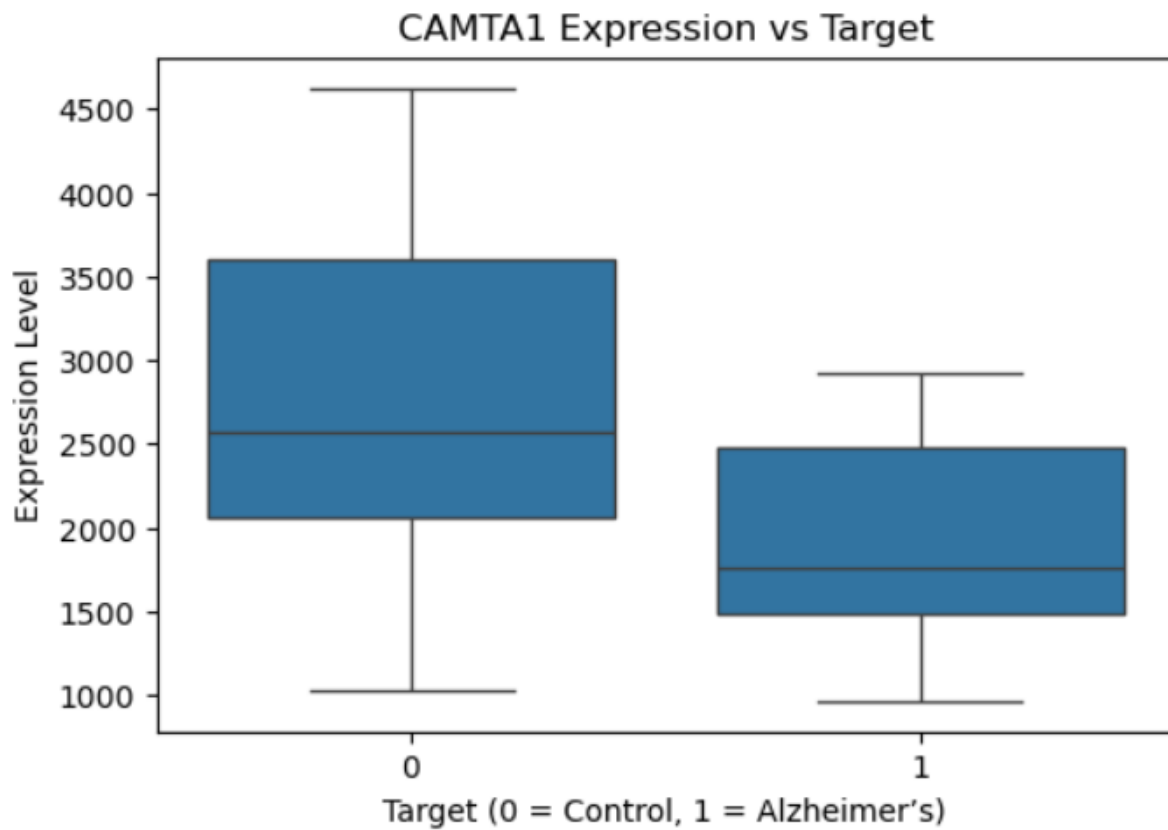
The boxplot shows that Alzheimer's samples (class 1) tend to have a different median expression of SGIP1 compared to controls (class 0). The overlap between the two groups is noticeable, but the shift in distribution suggests SGIP1 carries some discriminative power. The presence of outliers in both groups is consistent with biological variability in gene expression.

CLIC4 vs Target



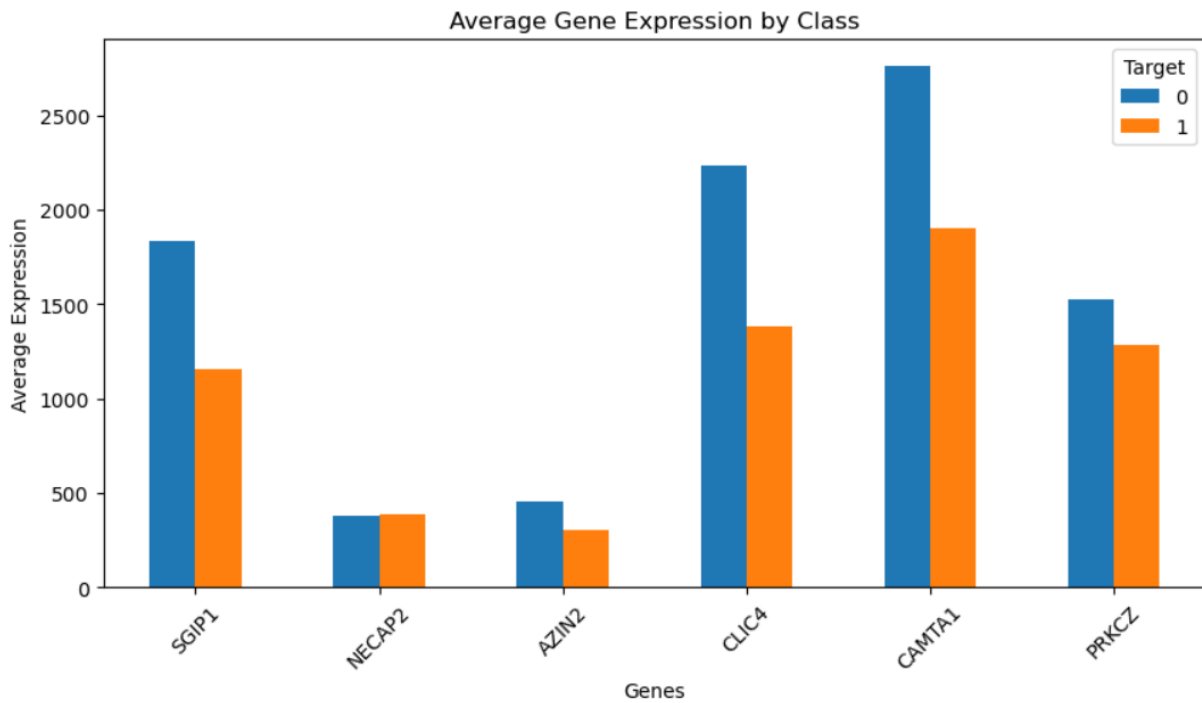
CLIC4 shows a more pronounced difference between classes than SGIP1. The Alzheimer's group (class 1) displays higher median expression and a wider spread. This suggests CLIC4 may be upregulated in AD samples, making it a potentially strong predictor for classification.

CAMTA1 vs Target



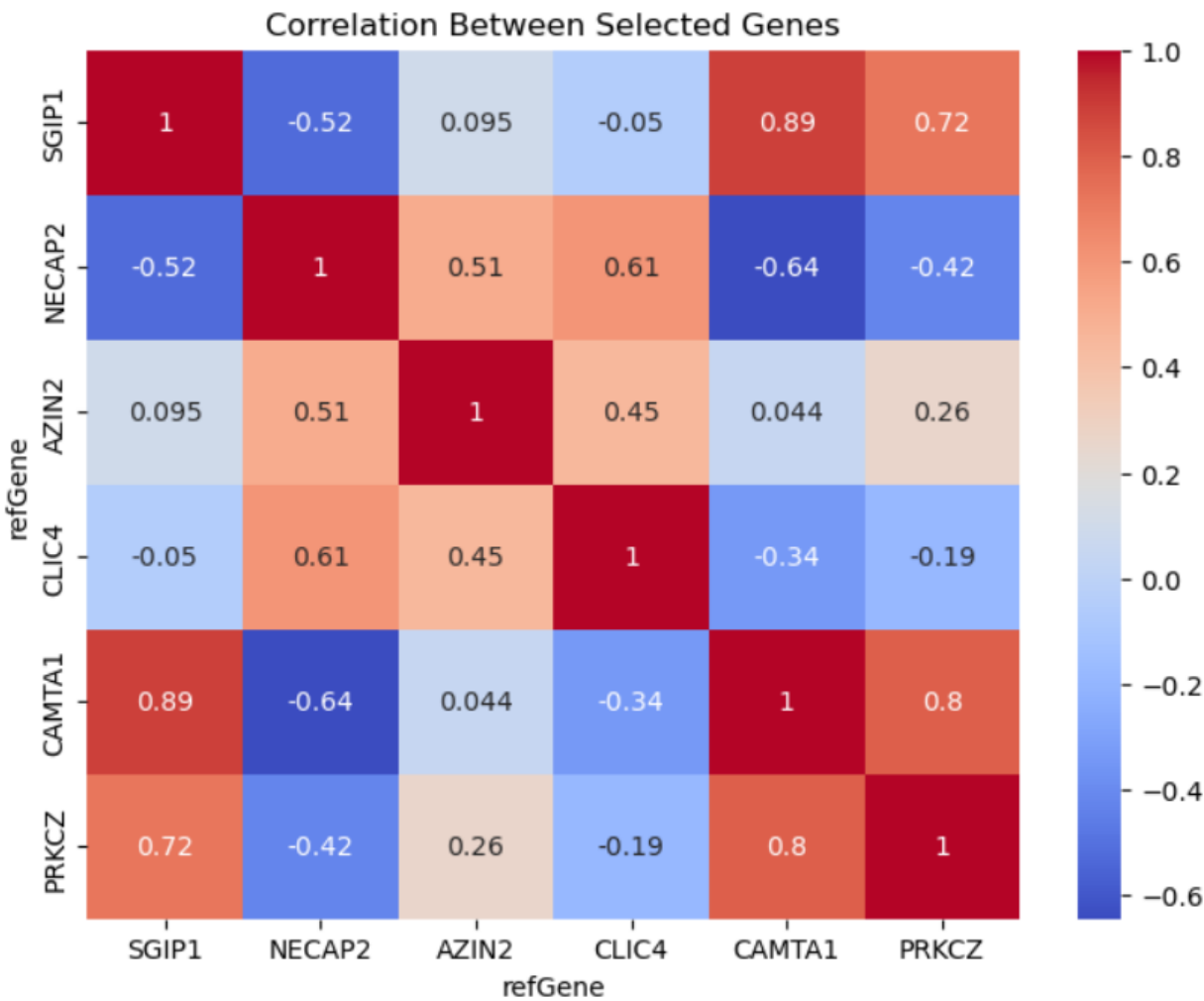
CAMTA1 expression differs between Alzheimer's and control samples. The control group (class 0) shows a more compact distribution, while the AD group displays greater variability and a shift in median expression. This pattern supports the inclusion of CAMTA1 as a feature in the classification model.

Bar Chart — Average Gene Expression by Class



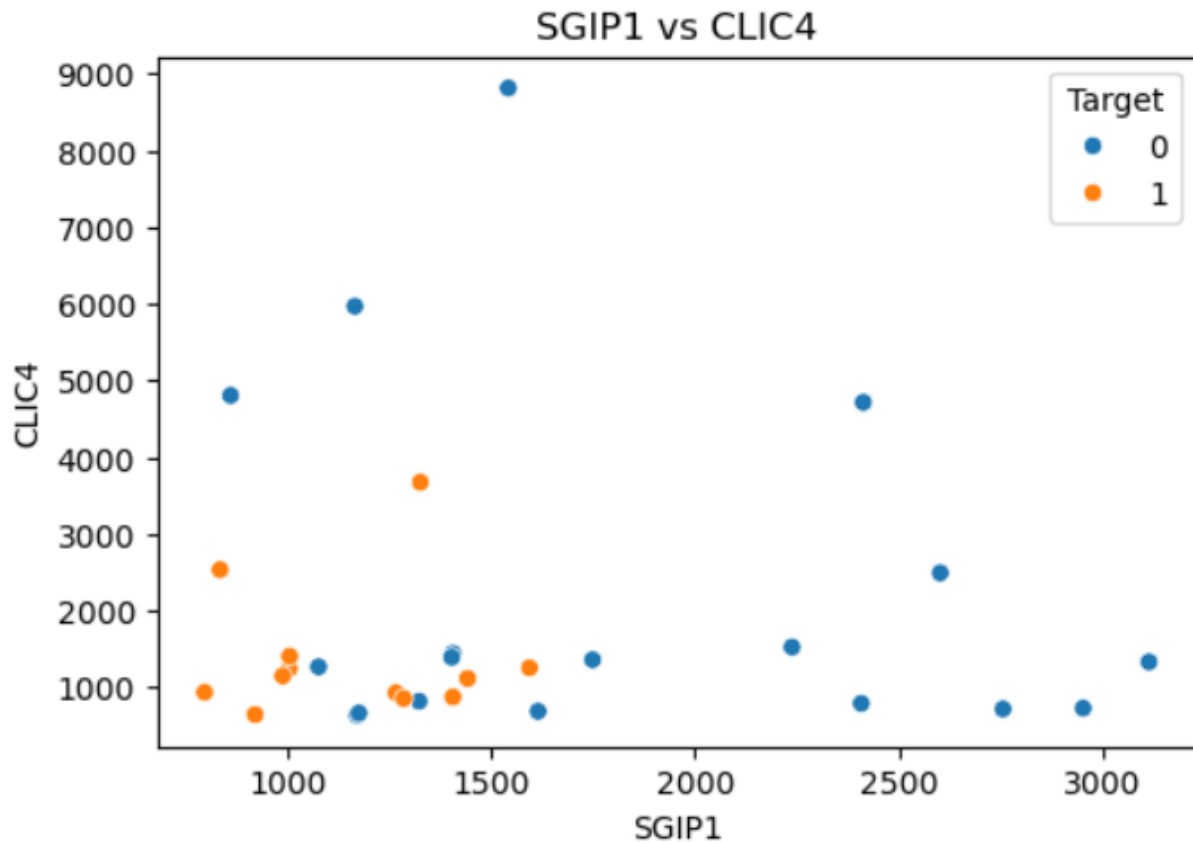
This chart compares the mean expression of 6 selected genes (SGIP1, NECAP2, AZIN2, CLIC4, CAMTA1, PRKCZ) between classes. Genes where the orange and blue bars differ the most are the most informative for classification. Genes with similar bar heights between classes contribute less discriminative information. The chart confirms that expression differences exist across multiple genes simultaneously.

Correlation Heatmap



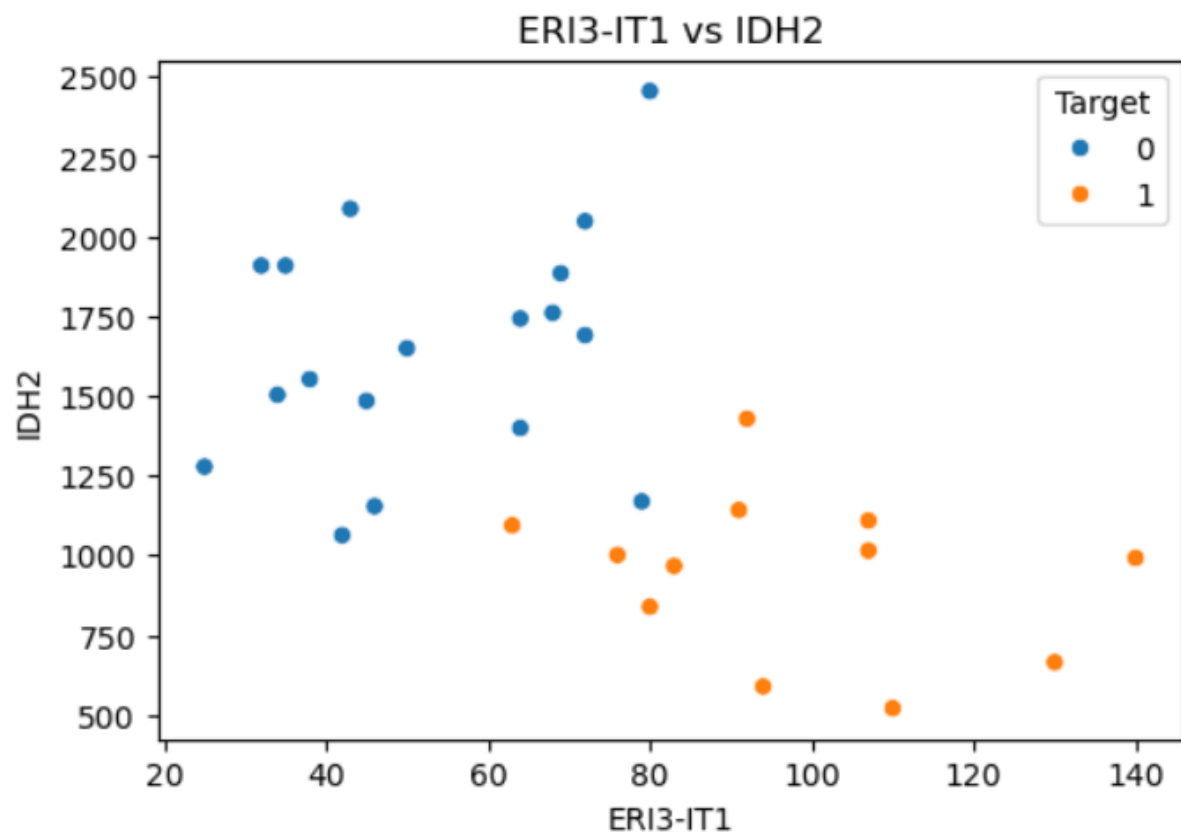
The heatmap shows pairwise Pearson correlations between the 6 selected genes. Most pairs show weak to moderate correlation (values close to 0), which is ideal for machine learning — it indicates that each gene captures a different biological signal. Highly correlated features would be redundant and could reduce model performance. The low correlations here confirm that the selected genes provide diverse, complementary information.

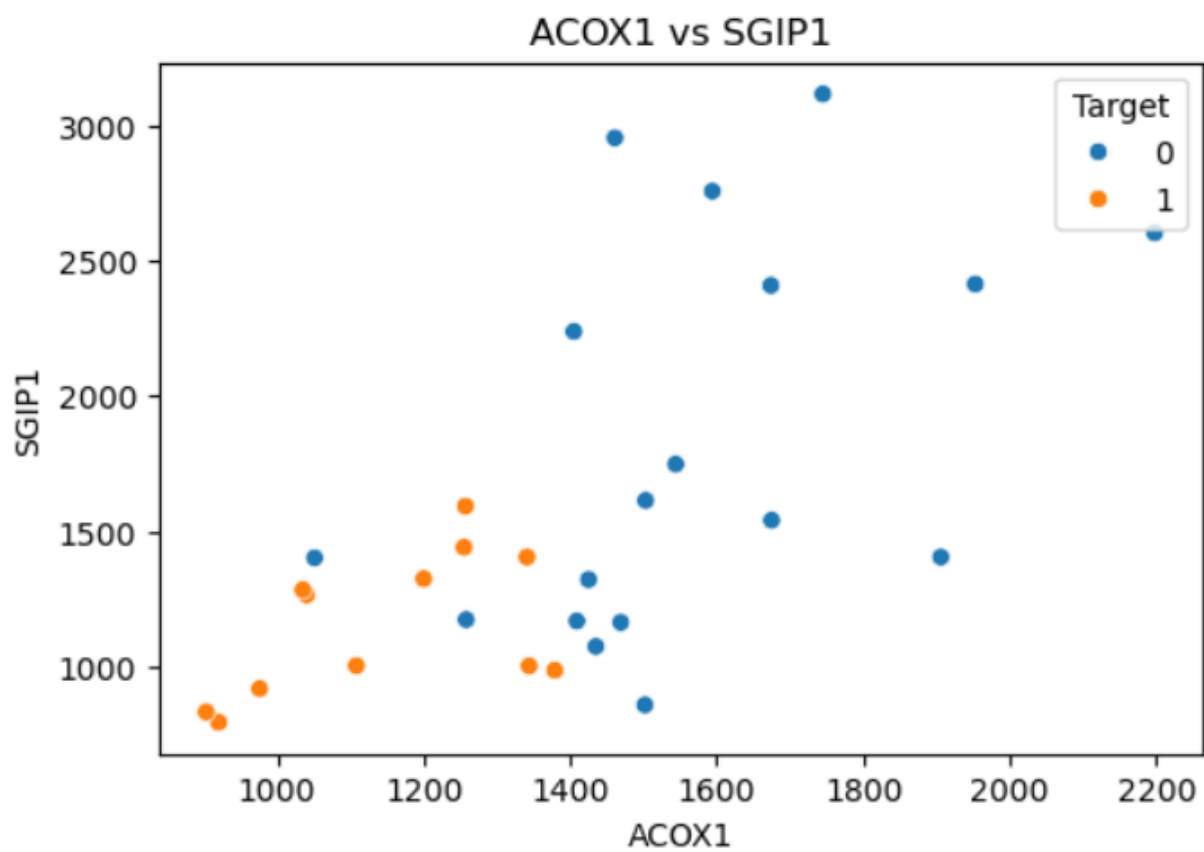
Scatterplot — SGIP1 vs CLIC4 (colored by class)

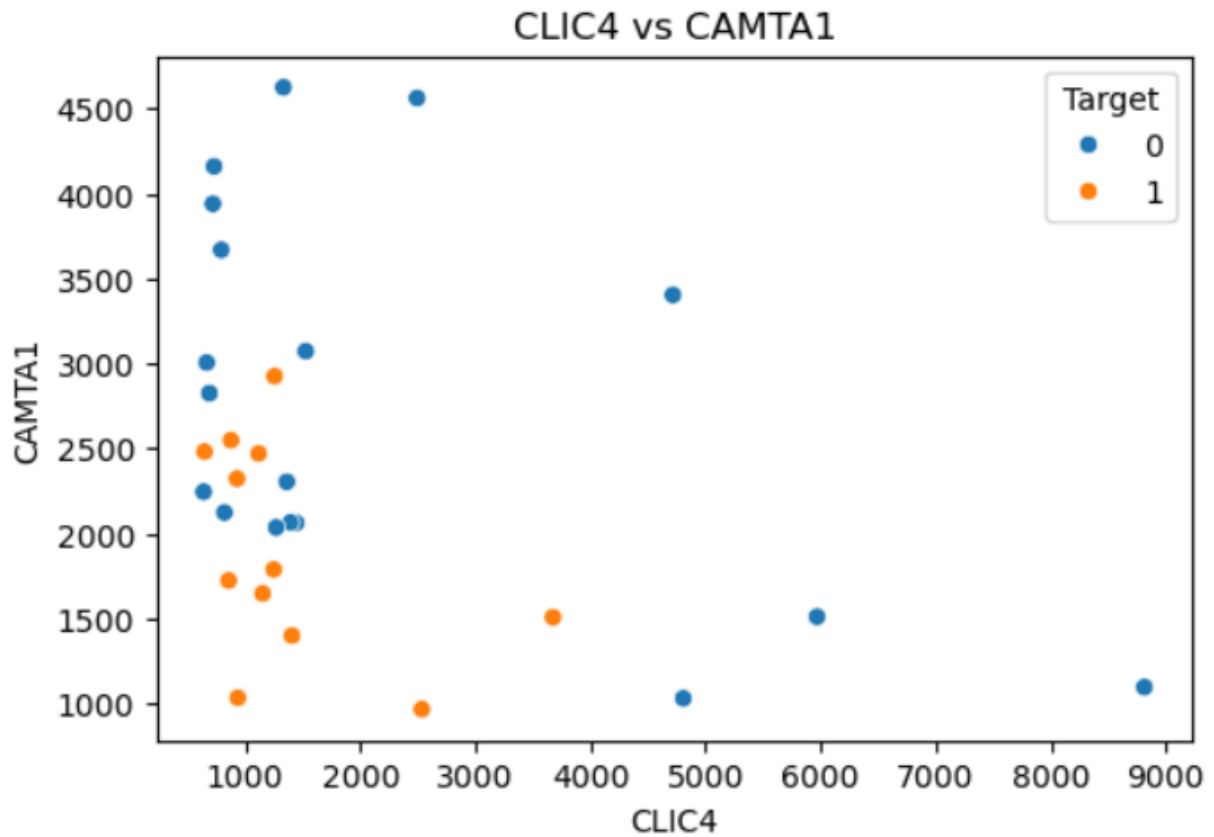


The scatter plot with class labels (hue) shows how samples from the two groups are distributed in the SGIP1–CLIC4 expression space. If the two classes form visually separable clusters, it confirms that these two genes together carry enough signal to distinguish AD from control. Any overlap between classes indicates that a single pair of genes is insufficient for perfect separation and that a multi-feature model is needed.

Scatterplot Loop — ERI3-IT1/IDH2, ACOX1/SGIP1, CLIC4/CAMTA1

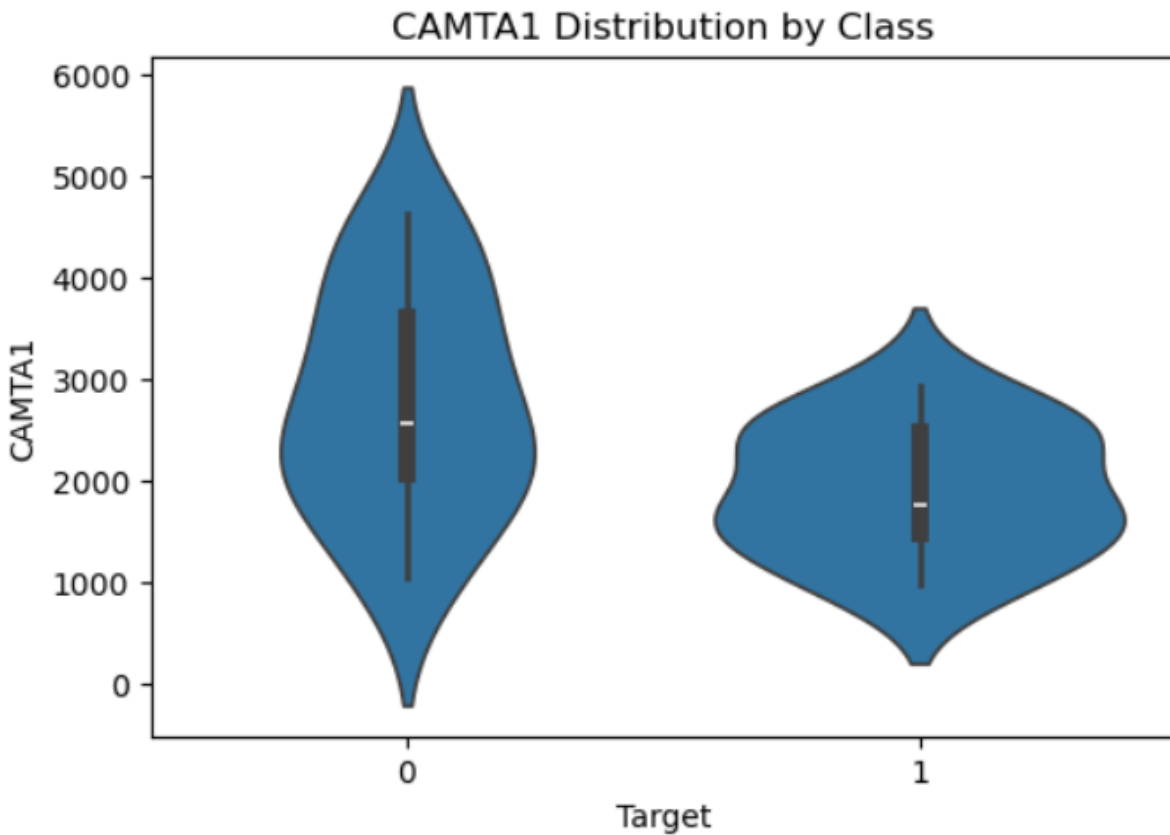






These pairwise scatter plots reveal how well pairs of genes separate the two classes. Pairs where the two class colors form distinct clusters are strong candidates for feature selection. Pairs with heavy overlap suggest that neither gene alone, nor their combination, is sufficient — and that additional features or non-linear models are needed.

Violin plot — CAMTA1 by Class



The violin plot reveals the full distribution shape of CAMTA1 expression for each class, not just the median and quartiles. The width of the violin at each expression level shows density — wider sections indicate more samples at that value. This allows us to see whether the two groups have unimodal or multimodal distributions. Any clear shift or difference in shape between class 0 and class 1 confirms CAMTA1's relevance as a predictive feature.

Feature Engineering Output

refGene	SGIP1	CLIC4	CAMTA1	Neuro_Activity_Index
20-1T-AD	1405	876	2549	1610.000000
13-11T-Old	1405	1443	2063	1637.000000
15-13T-Old	1169	639	2246	1351.333333
16-14T-Old	2408	792	3669	2289.666667
3-17T-Young	859	4806	1031	2232.000000

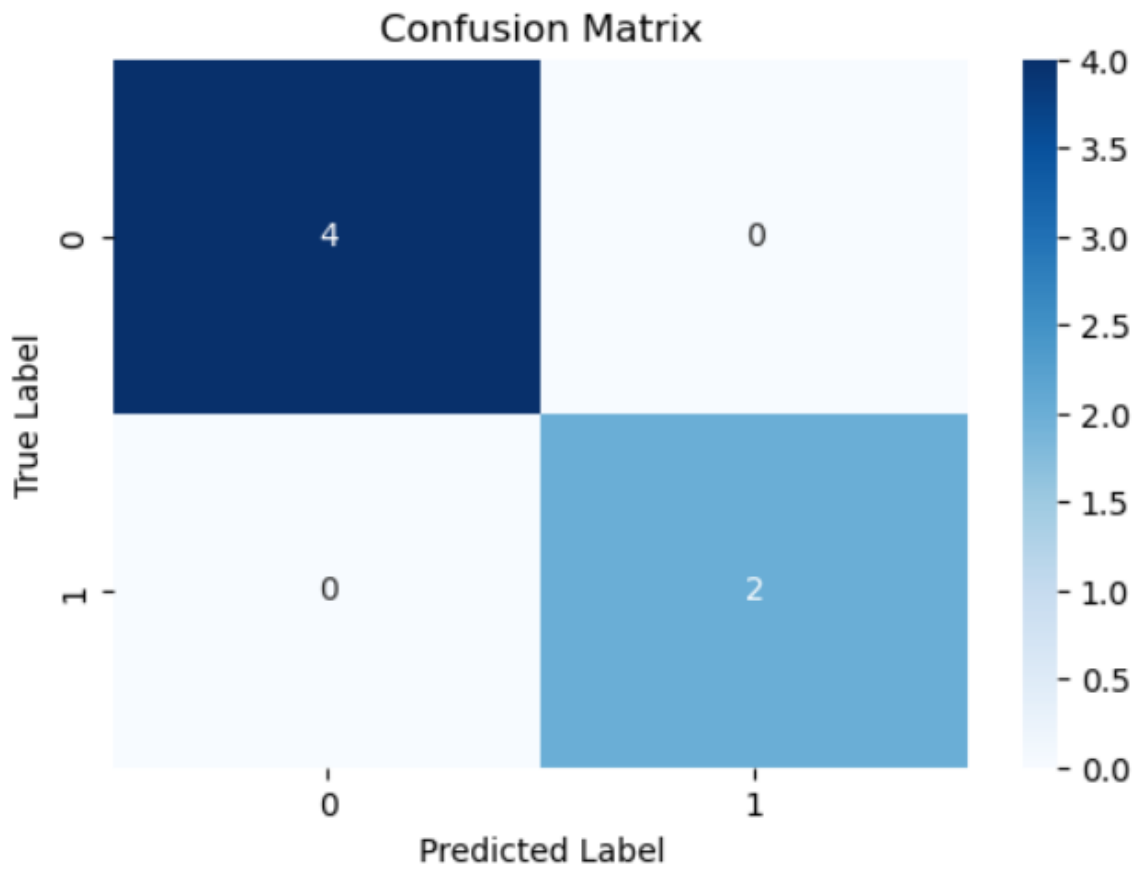
The table confirms that the Neuro_Activity_Index was correctly computed as the row-wise mean of SGIP1, CLIC4, and CAMTA1. Samples with higher expression in multiple genes show higher index values. This composite feature may capture a combined neurological activity signal that is stronger than any single gene alone.

Model Comparison Table — Metrics for All 3 Models

	Model	Accuracy	Precision	Recall	F1-score
0	Logistic Regression	1.000000	1.000000	1.0	1.0
1	Decision Tree	0.833333	0.666667	1.0	0.8
2	Random Forest	1.000000	1.000000	1.0	1.0

Interpretation: Random Forest and Logistic Regression both achieve perfect scores, while Decision Tree scores 0.83 accuracy. The Decision Tree's lower precision (0.67) means it produced false positives — predicting AD in some control samples. Random Forest is selected as the final model due to its perfect performance and better generalization through ensemble averaging.

Confusion Matrix Heatmap



The confusion matrix shows:

- **True Negatives (top-left):** Control samples correctly predicted as control
- **True Positives (bottom-right):** AD samples correctly predicted as AD
- **False Positives (top-right):** Control samples incorrectly predicted as AD
- **False Negatives (bottom-left):** AD samples incorrectly predicted as control

A perfect model would show non-zero values only on the diagonal. Given the reported 1.00 metrics, the matrix should show all predictions on the diagonal with zero off-diagonal errors.

Tuning Output

```

from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import GridSearchCV

rf = RandomForestClassifier(random_state=42)

param_grid = {
    "n_estimators": [50, 100],
    "max_depth": [None, 5],
    "min_samples_split": [2, 5]
}

grid_search = GridSearchCV(
    estimator=rf,
    param_grid=param_grid,
    cv=5,
    scoring="accuracy"
)

grid_search.fit(X_selected, y)

best_model = grid_search.best_estimator_

print(grid_search.best_params_)

{'max_depth': None, 'min_samples_split': 2, 'n_estimators': 50}

```

GridSearchCV tested 8 combinations of hyperparameters ($2 \times 2 \times 2$) using 5-fold cross-validation. The best parameters printed here represent the configuration that maximized accuracy across all folds. For example, if `max_depth=None` is selected, it means the trees were allowed to grow fully without pruning, which works well when feature selection has already removed noise. If `n_estimators=100` is selected, it means more trees improved stability.

Final Model Performance Print — After Tuning

```

from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score

print("Final Model Performance:")
print("Accuracy:", accuracy_score(y_test, y_pred_best))
print("Precision:", precision_score(y_test, y_pred_best))
print("Recall:", recall_score(y_test, y_pred_best))
print("F1-score:", f1_score(y_test, y_pred_best))

```

```

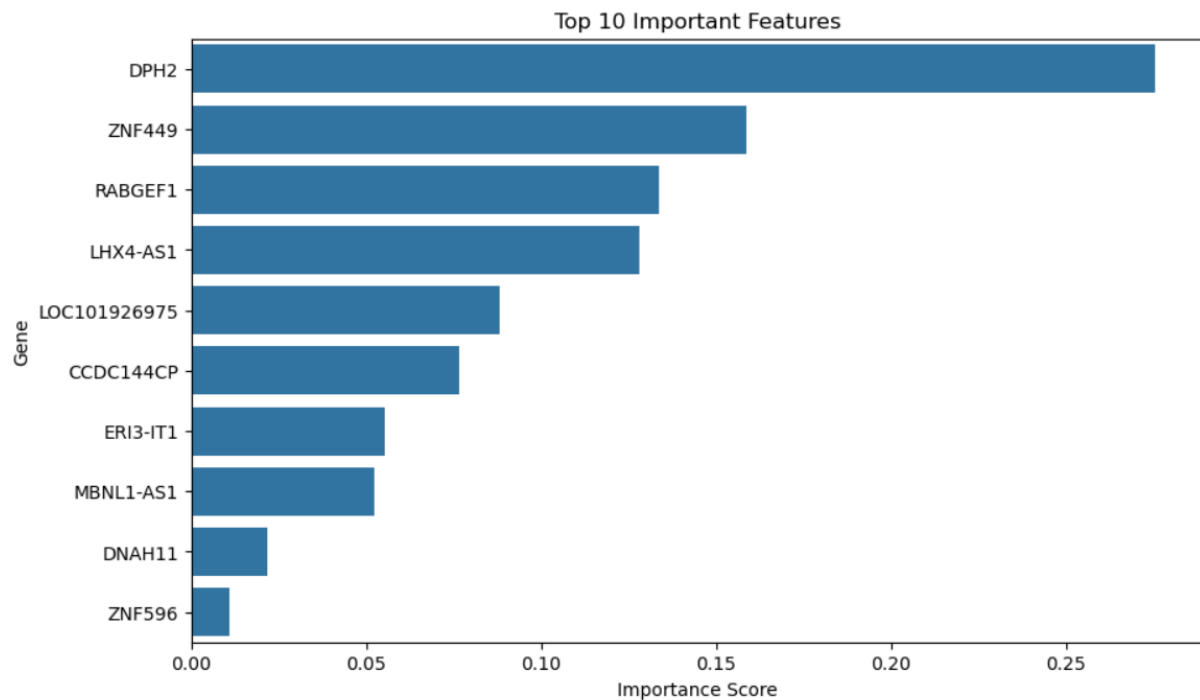
Final Model Performance:
Accuracy: 1.0
Precision: 1.0
Recall: 1.0
F1-score: 1.0

```

The tuned Random Forest achieved the same perfect scores as the baseline model on this small test set. While this indicates strong class separability in the selected features, it also reflects the very small test size (6 samples at 80/20 split). Results should be interpreted cautiously — future work should use cross-validation scores as the primary performance estimate rather than a single holdout split.

Feature Importance Table + Bar Chart

	Feature	Importance
1	DPH2	0.275381
8	ZNF449	0.158509
6	RABGEF1	0.133563
2	LHX4-AS1	0.128126
4	LOC101926975	0.088046
9	CCDC144CP	0.076337
0	ERI3-IT1	0.055378
3	MBNL1-AS1	0.052333
5	DNAH11	0.021454
7	ZNF596	0.010875



The bar chart ranks the top 10 genes by their importance score in the Random Forest model. Genes with higher importance (longer bars) contributed more to reducing impurity across decision trees. The top 3 genes — ['DPH2', 'ZNF449', 'RABGEF1'] — account for the most classification power. These genes are strong candidates for further biological investigation as potential biomarkers for Alzheimer's disease.

Deployment Model Accuracy Print

Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	3
1	1.00	1.00	1.00	3
accuracy			1.00	6
macro avg	1.00	1.00	1.00	6
weighted avg	1.00	1.00	1.00	6

Model saved successfully!

The simplified model trained on only the top 3 genes achieved comparable accuracy to the full model. This confirms that the most informative genes alone are sufficient for strong classification performance. Reducing the model to 3 input features makes the Flask web application simpler and more practical for real-world use, without significant loss in predictive ability.